EvidenceVault: StreamFold / dOCF: Run Manifests and Authenticated Run Views

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Abstract

Portable recursive-run evidence in this family is carried by a signed run manifest and authenticated scoped run views on one realized run scope. The manifest fixes replayable blob inventory together with the family's soundness and orchestration coordinates, and the views carry surfaced run claims on that same manifested run.

1 Run manifests and authenticated run views

Definition 1 (Run manifest)**.** *A run manifest is the family's manifest carrier M_{run} on one fixed run scope σ_{run}, where σ_{run} is the realized statement identifier, public inputs, and any carried local scope binding; write m_{run} for the canonical manifest name of that carrier on the same scope. The carrier M_{run} is signed and satisfies:*

(i) *it content-addresses the proved statement and public inputs;*

(ii) *it names every replayable blob required for deterministic replay;*

(iii) *it declares all soundness-relevant parameters and assumptions such as field sizes, repetition factors, independence tags, retry budgets, and aggregation-budget rules. For the common one-scalar random-linear-combination check, the residual failure term is at most $(n-1)/|\mathbb{F}|$ together with the usual $1/|\mathbb{F}|$ glue floor; t fresh independent repetitions or independently sampled scalars reduce that term to at most $((n-1)/|\mathbb{F}|)^t$, typically behaving like $1/|\mathbb{F}|^t$. The manifest must therefore carry the exact field, weight family, and repetition count actually consumed by the realized aggregation rule;*

(iv) *it carries one manifest-internal tagged entry family L of run-manifest entries $\Lambda = (q, p)$, where $q \in \{\mathsf{lineage}, \mathsf{ratchet}, \mathsf{override}, \mathsf{rollback}, \mathsf{exception}\}$ is the carried run-manifest entry kind and p is the corresponding payload. If $q = \mathsf{lineage}$, then $p = \ell$ and ℓ records the trigger, the parent attempt or stage identifier, the consumed budget slot or declared retry cap, what randomness, assignment, batching, or market-visible task changed, any coordinator action and resampling policy when applicable, the before/after checkpoint hashes, the resulting run-fragment or proof-task artifact digest, and any binding context needed to stop retry-budget laundering across sessions. If $q \in \{\mathsf{ratchet}, \mathsf{override}, \mathsf{rollback}, \mathsf{exception}\}$, then $p = \Gamma$, whose local δ-reading here is the affected run-stage coordinates together with any successor or rollback target digest and any declared exception or resampling semantics. Any realized ratchet, scoped override, rollback action, or exception activation therefore appears directly as one such entry in L rather than through a second family-local governance surface.*

(v) *it records the shard-partition rule, batching and aggregation plan, prover role assignments, any allowed substitution surface for market participants, and any carried coordinator references needed to replay the realized run.*

Definition 2 (Authenticated scoped run view)**.** *An authenticated scoped run view is the family's replay-facing carrier on that fixed run scope and gives the recursive-pipeline family's local portable run claim*

$$\mathsf{RV} = (\sigma_{\mathrm{run}}, m_{\mathrm{run}}, \Sigma, \psi, f, A),$$

where σ_{run} is the family's run scope, m_{run} names the canonical run manifest carrier M_{run} on that same scope, and Σ is the carried bridge replay slice for that manifested run, ψ is the carried replay-facing role package, with $k \in \{\mathsf{full}, \mathsf{front}\}$ selecting the local verification view, f is the carried payload for that bound role, and A is the scoped authority evidence for any signer, submitter, coordinator, or delegation chain named by the claim. Deployment often uses a fast front and a fuller audit surface, but those are only the two local readings of k on one realized run scope. Portability therefore still requires deterministic verification and transcript reconstruction to refine the same manifested run, and any claimed fuller surface must carry whatever conformance material or differential witness explains a mismatch with a claimed fast front.